AI-Powered Data Analysis Remote Internship

Week 1 Deliverable

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Understanding the Dataset Report

Purpose and Scope

This report explores the Student_Opportunity_Participation dataset with 8,558 records, 22 opportunities, 5 opportunity categories drawn from 70 countries and 2,089 institutions. Date coverage varies by field: Apply Date spans October 2022 to April 2024, while Learner Signup Date Time spans January 2023 to April 2024; the Historical Analysis section below uses Apply Date, since it has the fullest coverage. The dataset captures the full participation lifecycle; from sign-up and application through allocation, completion, or exit across five opportunity categories.

The report gives a detailed overview of the dataset, revealing an understanding of the data through the data dictionary, a quality assessment report, initial analysis and visualizations to capture historical trends and performance of each Opportunity Category.

1 Dataset Overview

The Student Participation Dataset contains information about students and their involvement in a range of academic, professional, and engagement-related opportunities. The dataset consists of 8,558 participation records and 16 variables. Each row represents a learner–opportunity participation record, linking a student with a specific opportunity and the corresponding participation status.

The dataset combines student-related information with details about the opportunities available to them. This makes it possible to understand not only what types of opportunities students participate in, but also how their participation progresses from initial registration or application to different participation outcomes.

Overview of the Student Participation Dataset

Dataset Characteristic	Value
Total records	8,558
Total variables	16
Unique Opportunity IDs	23
Distinct Opportunity Names	22
Opportunity Categories	5
Participation Status Types	8

The difference between the number of Opportunity IDs and distinct Opportunity Names indicates that one opportunity name may be associated with more than one identifier. This should be considered when opportunities are compared in later stages of the analysis.

1.1 Types of Opportunities

The dataset includes opportunities from five main categories:

- Internship
- Course
- Event
- Competition
- Engagement

These categories represent different forms of student development and participation. Internships primarily relate to professional and workplace experience, while courses support learning and skill development. Events and competitions provide students with opportunities for interaction, exposure, and achievement, whereas engagement opportunities capture other forms of student involvement.

1.2 Information Captured in the Dataset

The 16 variables can be broadly divided into four groups: student information, opportunity information, participation information, and date/time information.

Main Types of Information Recorded

Information Group	Variables / Examples
Student information	First Name, Date of Birth, Gender, Country, Institution Name, Current/Intended Major
Opportunity information	Opportunity ID, Opportunity Name, Opportunity Category
Participation information	Status Description, Status Code
Date and time information	Learner Signup Date Time, Apply Date, Opportunity Start Date, Opportunity End Date, Entry Created At

Student-related variables provide demographic and academic context, including gender, country, institution, and intended or current major. Opportunity-related variables identify the specific opportunity and the category to which it belongs. The dataset therefore allows participation to be examined in relation to both student characteristics and the type of opportunity selected.

1.3 Recording Student Participation and Engagement

Student participation is mainly captured through the Status Description and Status Code variables. The dataset contains eight different participation-status descriptions:

Applied, Started, Team Allocated, Waitlisted, Rejected, Withdraw, Dropped Out, and Rewards Award.

These statuses represent different stages or outcomes of a student's interaction with an opportunity. For example, an Applied status indicates that a student has submitted an application, while Started indicates progression into the opportunity. Team Allocated reflects placement into a team-based activity, whereas Dropped Out and Withdraw indicate that participation did not continue. Rewards Award represents successful recognition or completion-related achievement. The dataset does not contain a separate numerical engagement score. Instead, student engagement is represented indirectly through participation status and progression across different stages of an opportunity, supported by registration, application, and opportunity date information.

Participation is also supported by several date and timestamp variables. Learner Signup Date Time records when a learner registered, Apply Date indicates when an application was made, and the opportunity start and end dates describe the period during which the opportunity takes place. Together, these variables provide a chronological view of student participation.

Data Dictionary

A description of the variables, the datatypes and their analytical relevance. Click to access the data dictionary: (Student_Participation_DataDictionary)

Table: Data Dictionary — Student Participation Dataset (16 variables)

Variable	Description	Data Type
Learner Signup Date Time	Date and time the learner registered on the platform	Datetime (text)
Opportunity Id	Unique identifier for the specific opportunity instance	Text (ID string)
Opportunity Name	Name/title of the opportunity	Text
Opportunity Category	Category the opportunity belongs to (5 values: Internship, Course, Event, Competition, Engagement)	Categorical (text)
Opportunity Start Date	Date/time the opportunity begins	Datetime (text); 44% missing
Opportunity End Date	Date/time the opportunity ends	Datetime (text)
First Name	Learner's first name	Text

Variable	Description	Data Type
Date of Birth	Learner's date of birth	Date (text)
Gender	Learner's reported gender (4 categories)	Categorical (text)
Country	Learner's country of residence	Categorical (text); 70 distinct after cleaning
Institution Name	Learner's academic institution	Text; 5 missing values
Current/Intended Major	Learner's field of study	Text; 5 missing values
Entry created at	Timestamp the database record was created	Datetime (text)
Status Description	Participation stage or outcome (8 values: Applied, Started, Team Allocated, Waitlisted, Rejected, Withdraw, Dropped Out, Rewards Award)	Categorical (text)
Status Code	Numeric code mapped 1:1 to Status Description	Integer (1010–1120)
Apply Date	Date/time the learner applied to the opportunity	Datetime (text)

Data Quality Report

Severe exploration was done for the dataset containing 8,558 records and 16 variables covering learner information, opportunity information, application dates, and application status.

Each section below describes the Issues Found and how they have been addressed.

Data Cleaning Overview			
Column	**Issues Found**	**Insights**	**Handling Measures**
Learner Sign Up Date Time	295 of 8,558 values failed standard parsing. 2 corruption patterns in the time-of-day portion: hour written as 24 instead of 00 (midnight), and a large spurious number glued onto the hour (e.g. '1089960:00:00').	The corrupted time values are identical across all rows sharing the same date, this is a deterministic export bug, not real data. The date (month/day/year) itself is valid in 100% of rows.	Extracted the date separately from the time, and normalized the hour 24 to 00:00:00 on the same day. Flagged these rows in a new Boolean column, "Signup time corrupted"
Opportunity category	Categorical field. Checked for stray whitespace, casing variants, and unexpected categories.	No casting or typo variants found	Stripped any leading/trailing whitespaces.

Data Cleaning Overview			
Opportunity Start Datetime	3794 of 8558 values (44%) are blank. Time glue corruption pattern for another 843 rows	Not necessarily a corrupted date, but a missing date.	Left blanks as NaT. Normalized the corrupted time-glue onto the hour as 00:00:00, leaving the dates as they are
Opportunity End Date	1262 unparseable time values with time glued on hour values, and 24hr instead of 00.	Perform cleaning similar to the 'learner sign up date', flagging the corrupted time in a new Boolean column 'opportunity_end_time _corrupted'	Splitting the date from datetime. and normalizing the glued hour to 00:00:00
First Name	36 values appear like IDs or placeholders/test data and not real names of people: i.e. 'Manish3Test3', 'K', numbers like '11, 452-', ID like string (22C126 Surya deva). Inconsistent casing for Some name values.	Reserving the names because they still hold value in future participation analysis	Formatted the casing style to a proper casing.
Date of Birth	1 date of birth suggests that the learner's age is 5 yrs when applying for an opportunity.	A data entry error for the DOB, implicating an out-of-range value.	Kept the date, flagging as implausible
Country	1 country inconsistency in naming appearing twice: ie 'Iran, Islamic Republic of Persian Gulf' and 'Iran Islamic Republic of Persian Gulf'	The country is the same, seemingly a data-entry error	Standardized the name to merge as the same country.
Institution Name	5 missing values, casing duplication, same school name appearing many times but differently. ie 'Saint louis university', 'SAINT LOUIS UNIVERSITY'	Casing issues, whitespaces, trailing/heading	Normalized casing (trim, lowercase, collapse whitespace, title-case) to collapse pure case/spacing duplicates. Left missing values as NaN
Current/Intended Major	5 missing values, with near duplicate variants		Trimmed whitespaces only. reserved the variant as they are, and left missing values as

Data Cleaning Overview			
			NaN
Apply Date	Same time-corruption patterns (307 unparseable before cleaning). Additionally, 99 rows show an Apply Date earlier than the learner's own Signup Date, which is logically impossible	The apply date appearing earlier than the sign-up date may need more clarification in the case of analyzing days taken to apply for a program after sign-up	Applied datetime split and normalization same as in the other datetime columns with the same issue.

Data Quality Assessment Overview

A data quality assessment was carried out across five dimensions: **completeness, uniqueness, consistency, validity, and accuracy**, to determine the dataset's reliability for analysis.

1. **Completeness:** Most fields are highly complete. Opportunity Start Date is the main gap, with 3,794 missing values (44% of records). Institution Name and Current/Intended Major each have 5 missing values, a negligible share of the dataset.
2. **Uniqueness**: No fully duplicate records were found across all 16 fields, confirming the dataset is unique at the row level. Institution Name, however, showed duplication in substance rather than in whole rows; the same institution appeared under multiple casings (e.g. "Saint Louis University," "saint louis university," "SAINT LOUIS UNIVERSITY"), inflating the field from 2,089 to what is really 1,797 distinct institutions once casing is normalized.
3. **Consistency**: Structured categorical fields (Opportunity Category (5 values), Gender (4 values), and Status Description/Status Code) are internally consistent, with each Status Code mapping to exactly one Status Description across all records.
4. **Validity**: Date fields were checked for valid formats and logical sequencing. Learner Signup Date Time, Apply Date, and Opportunity End Date each contained a corrupted time-of-day component in roughly 3–15% of records (invalid hour values), though the calendar date itself was valid in all cases and was preserved. Logical sequencing checks also surfaced two issues, counted after recovering the date portion from corrupted timestamps: 99 records where the Apply Date precedes the learner's own Signup Date, and 116 records (of 4,764 with both dates present) where the Opportunity End Date precedes the Opportunity Start Date — both are impossible sequences that warrant review with the source system rather than automatic correction.
5. **Accuracy**: The dataset is broadly fit for analyzing learner historical participation and opportunity trends, but a small number of records show signs of test or placeholder data — 36 First Name values resemble IDs or dummy entries (e.g. "Manish3Test3") rather than real names — and 1 record has an implausible Date of Birth relative to signup age. These affect under 0.5% of records combined and have been flagged rather than removed.

6. **Overall Quality**: The dataset is suitable for analysis at scale, with the missing Opportunity Start Date values representing the most material limitation: any duration or timeline-based analysis shall be analyzed based on the date rather than date and time. The remaining issues (corrupted timestamps, casing duplication, illogical date ordering, and a handful of placeholder records) are minor in volume and have been corrected or flagged during cleaning; which should be noted as known limitations when interpreting downstream results.

Historical Analysis

Overall Trajectory in Application Trend: Applications were minimal through Oct 2022 to 2023, a sudden wave in Mid 2023 followed by an upward trend from November 2023 into 2024 with January 2024 hitting the peak volume of 2,400+ signups - roughly 150X more than the initial Oct 2022 volume.

Growth Trends and Patterns: The growth pattern in sign-ups/applications is not smooth, its cyclical with sharp spikes.

Initially, volume began modestly in early 2023 followed by a pullback in April, then climbed steadily through the spring into a first peak in August 2023.

A stronger spring sustained volume through autumn before surging again to the peak volume of 2,400+ sign-ups in January 2024, more than double the prior peak of 788 in August 2023.

A dramatic December 2023 towards January 2024 surge is by far the largest growth streak in the dataset.

The two-wave pattern (August 2023 and January 2024), consistent with participants enrolling around the start of academic terms, with each wave outperforming the last is a sign of program growth over the observed period.

Breaking the trend out by category opportunities, Internships drove almost all of the growth and both seasonal peaks— Internships are the primary engine of the program's historical growth.

Significant observations

A December-to-January spike is clearly visible in the Dec 2023 → Jan 2024 transition (682 → 2,488 applications). The same transition a year earlier (Dec 2022 → Jan 2023: 9 → 30) is far too small to call a spike, so this looks like a single large event tied to Internship launches rather than a confirmed recurring seasonal pattern; it's a hypothesis worth testing against a second full year of data, possibly aligned with new-year application cycles or a new opportunity cohort opening.

Growth is heavily event-driven rather than steady: ‘Internship’ opportunities attract the largest volume of signups - specific opportunity launches are significantly driving volume. This may imply that the Learners in the platform are attracted to the specific opportunities, not organic gradual adoption.

The first three months of data (Oct–Dec 2022) are thin enough that they may represent a soft-launch/pilot period for the Opportunities towards the full operation towards the end of 2023.

Opportunity Category Analysis

Distribution and Outcome Framework

Participation across the five opportunity categories is heavily skewed toward Internships, which make up 63.34% of all records (5,421), followed by Courses at 23.80% (2,037). Event, Competition, and Engagement together account for just 12.86% of participation. Volume alone, however, says little about how well participants actually progress — to assess that, the eight recorded statuses were grouped into three outcome classes: Positive/Progression (Started, Team Allocated, Rewards Award), Negative (Rejected, Dropped Out, Withdraw), and Pending/Intermediate (Applied, Waitlisted).

Outcomes by Category

Viewed across the whole dataset, outcomes look deceptively balanced — 49.92% Negative against 47.58% Positive, with only 2.50% still pending. That balance disappears once outcomes are broken out by category. Internship is the only category classified as a Bottleneck: just 21.49% of its records reach a Positive outcome, while 78.23% end Negative, driven mainly by Rejected (65.84%) and Dropped Out (11.31%) statuses. Every other category tells the opposite story — Course, Event, Competition, and Engagement all post Positive rates above 84%, with Course, Event, and Engagement exceeding 93%. In these categories, Team Allocated is consistently the dominant status, meaning positive outcomes come from allocation into a team rather than surviving a graded application process, which is the mechanism that drags Internship down. Competition is a partial exception, carrying the largest Pending share (14.59%) as more of its records sit at the Applied stage.

Category	Records	Positive %	Negative %	Pending %	Rating
Internship	5,421	21.49%	78.23%	0.28%	Bottleneck
Course	2,037	93.18%	1.18%	5.65%	Strong Performer
Event	545	96.33%	0.37%	3.30%	Strong Performer
Competition	425	84.24%	1.18%	14.59%	Strong Performer
Engagement	130	96.92%	0.00%	3.08%	Strong Performer

Volume vs. Performance

Benchmarking each category against the overall Positive (47.58%) and Negative (49.92%) rates confirms Internship as the single Bottleneck in the program, and notably the largest category by volume — so high participation does not translate into stronger outcomes here, if anything the reverse holds. Plotting volume against Positive rate makes this concrete: Internships sit in a high-volume, low-performance quadrant, while Courses show that strong performance is achievable even at meaningful scale (2,037 records, 93.18% Positive). Competition and Engagement post strong Positive rates too, but their small volumes limit how much they can offset the Internship shortfall overall.

Opportunity-Level Patterns

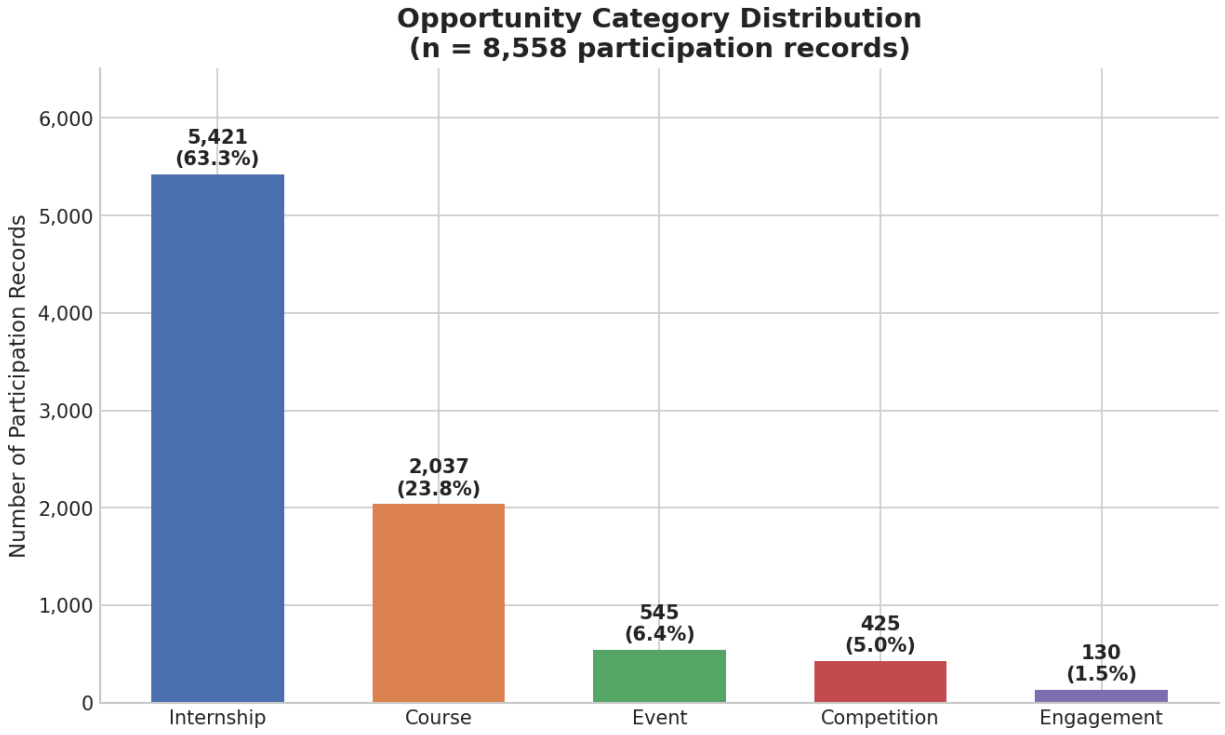
Digging into individual opportunities (restricted to those with ≥30 records, 20 of 22) shows this isn't

a case of one bad Internship dragging down an otherwise healthy category. Health Care Management is the best-performing Internship opportunity at 54.85% Positive, but every other Internship opportunity falls between roughly 11% and 22%. Removing Health Care Management from the Internship total still leaves 83.91% Negative — the problem is spread across the category, not concentrated in a few weak listings. At the same time, the strongest performers dataset-wide — Startup Mastery Workshop (Event, 99.24%), UX Redesign Challenge (Competition, 98.21%), and Urban Renew Challenge (Competition, 97.83%) — are all non-Internship opportunities, and no Internship opportunity appears anywhere in the top 10.

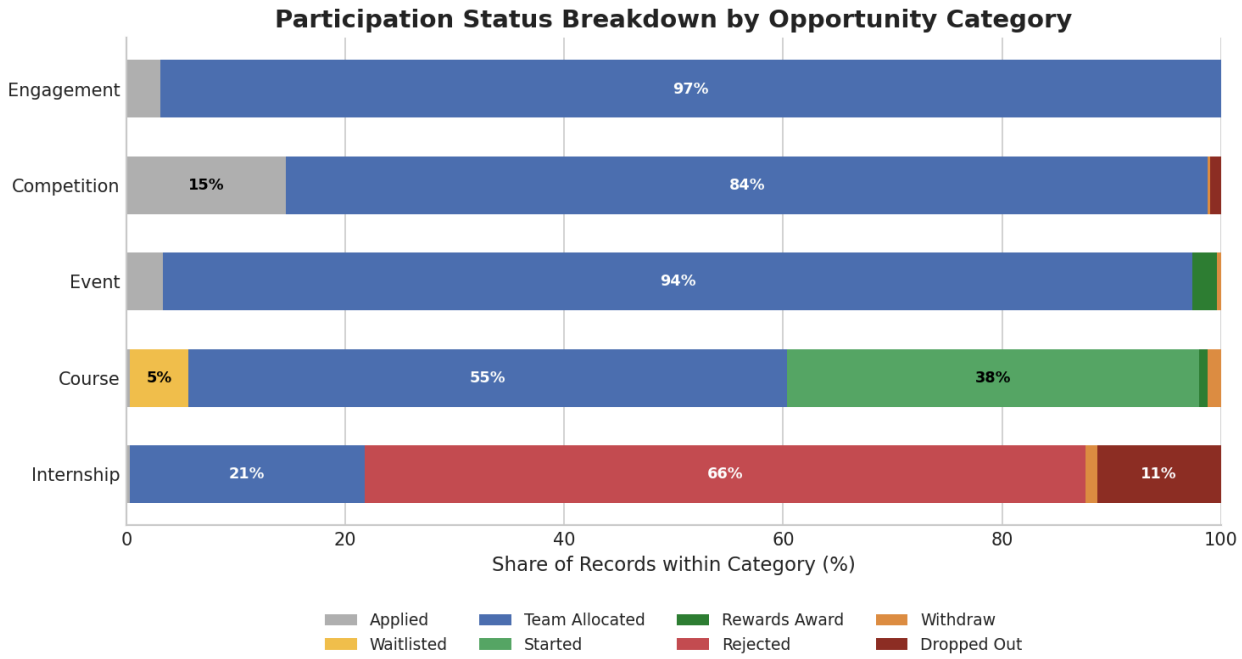
Visualizations

Category Charts

Category Distribution — confirms your reported split exactly: Internship 63.3%, Course 23.8%, Event 6.4%, Competition 5.0%, Engagement 1.5%.

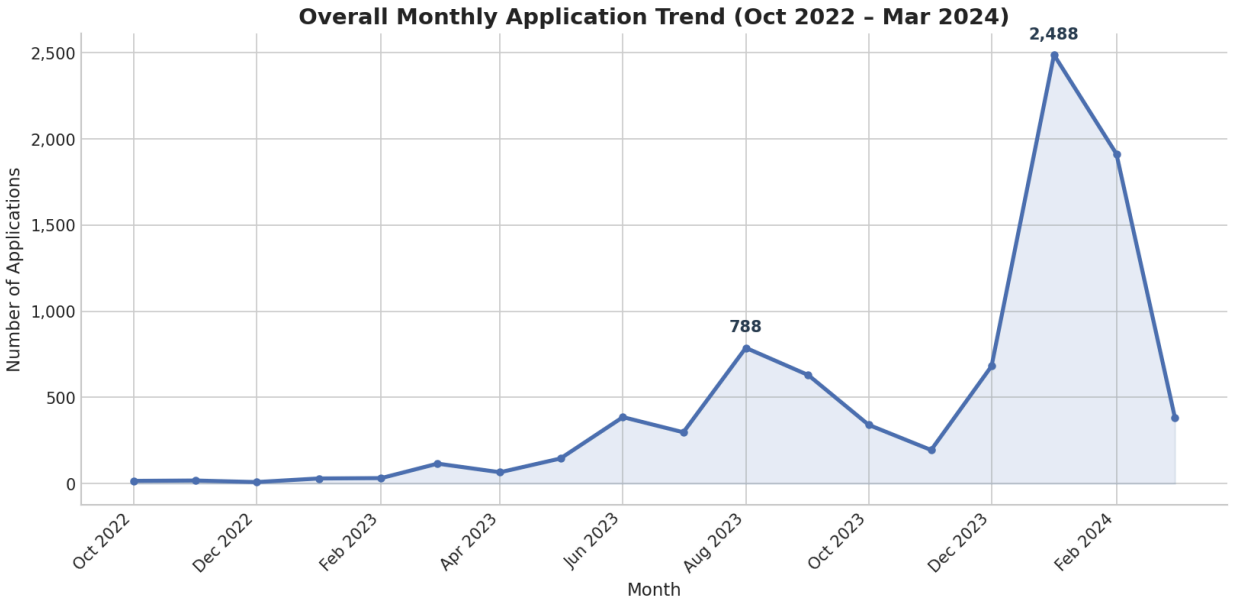


Status Breakdown by Category — a 100% stacked bar showing each category's status mix. This surfaces something worth calling out in your report: *Rejected* only appears in Internship (66% of Internship records), while Course is the only category with *Started*/*Rewards Award* statuses.

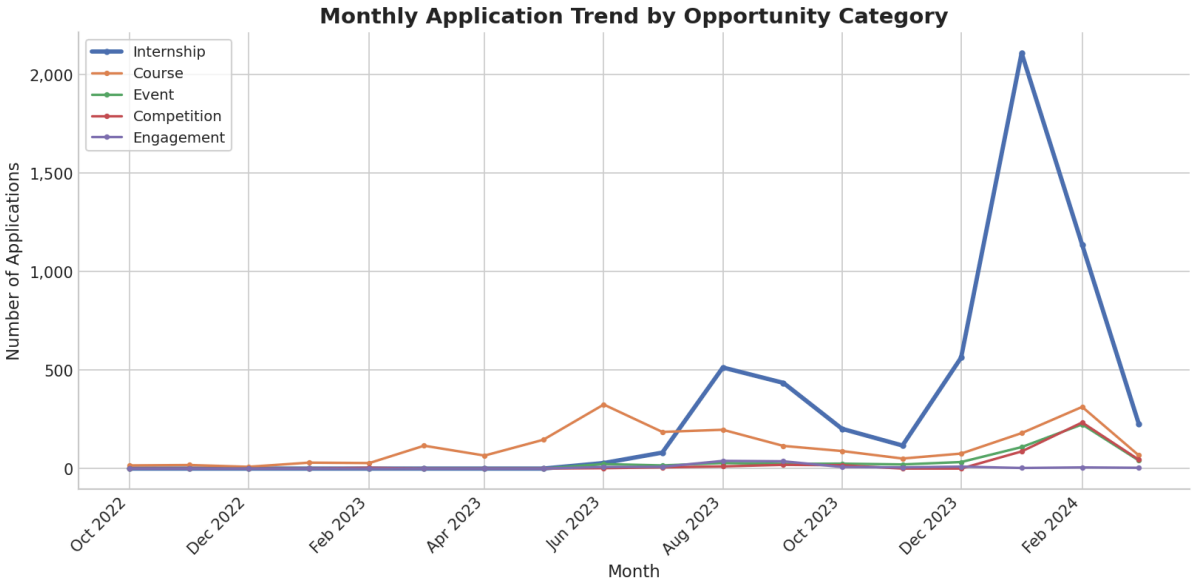


Trend Charts

Overall Monthly Applications — reproduces your two-wave pattern almost exactly (Aug 2023: 788, Jan 2024: 2,488), built from Apply Date since that's the field that actually starts in Oct 2022.

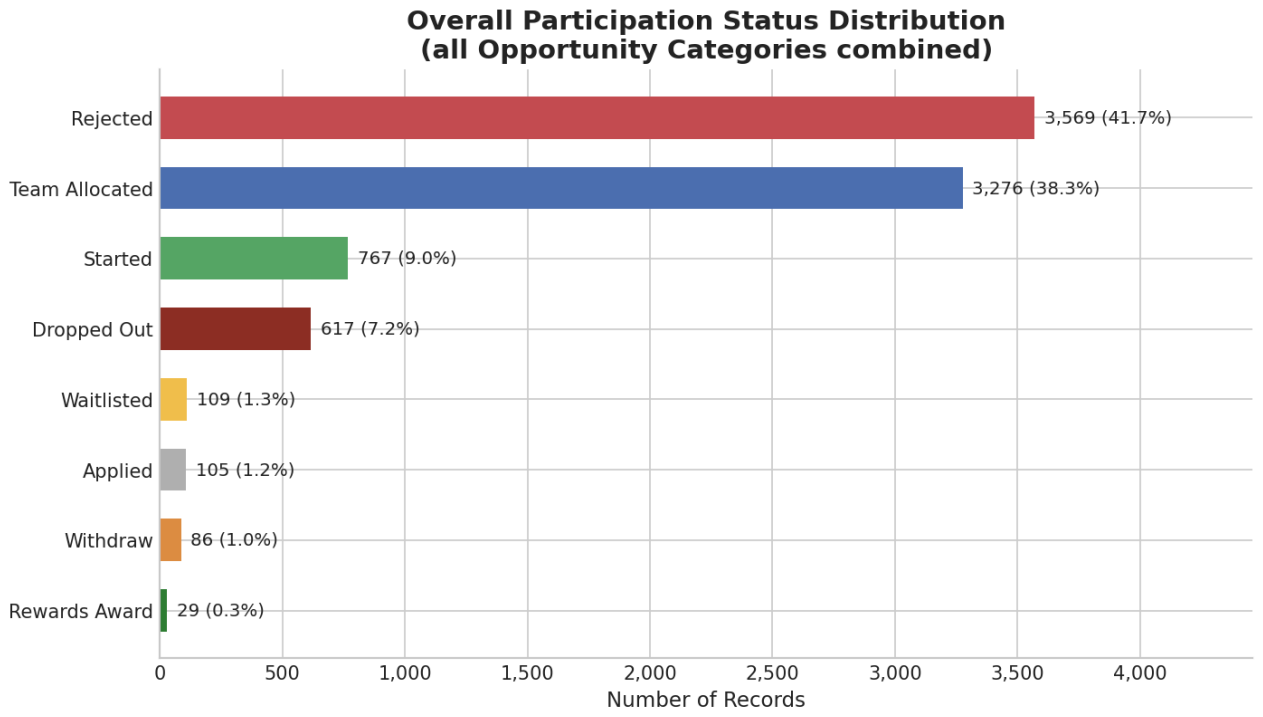


Trend by Category — shows Internship driving both peaks while other categories stay flat, backing up your "Internships are the primary engine of growth" claim.

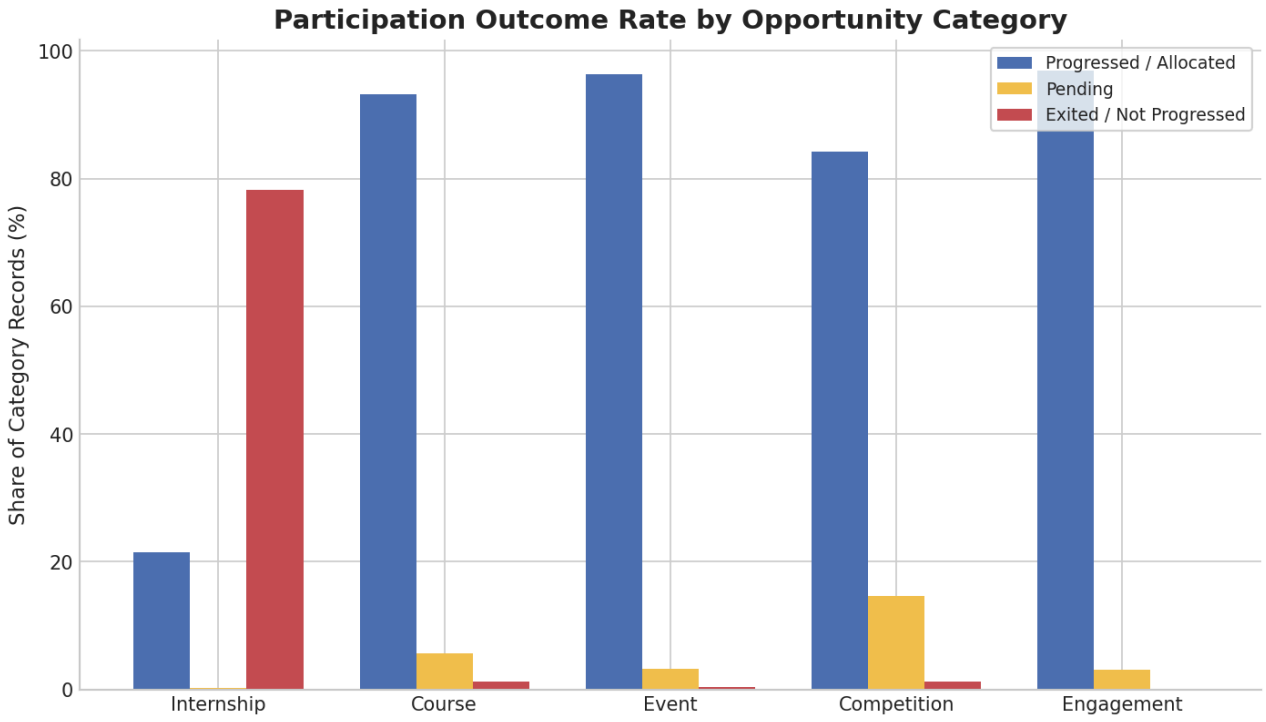


Summary graphs

Overall Status Distribution — Rejected (41.7%) and Team Allocated (38.3%) dominate; together they're 80% of all records.

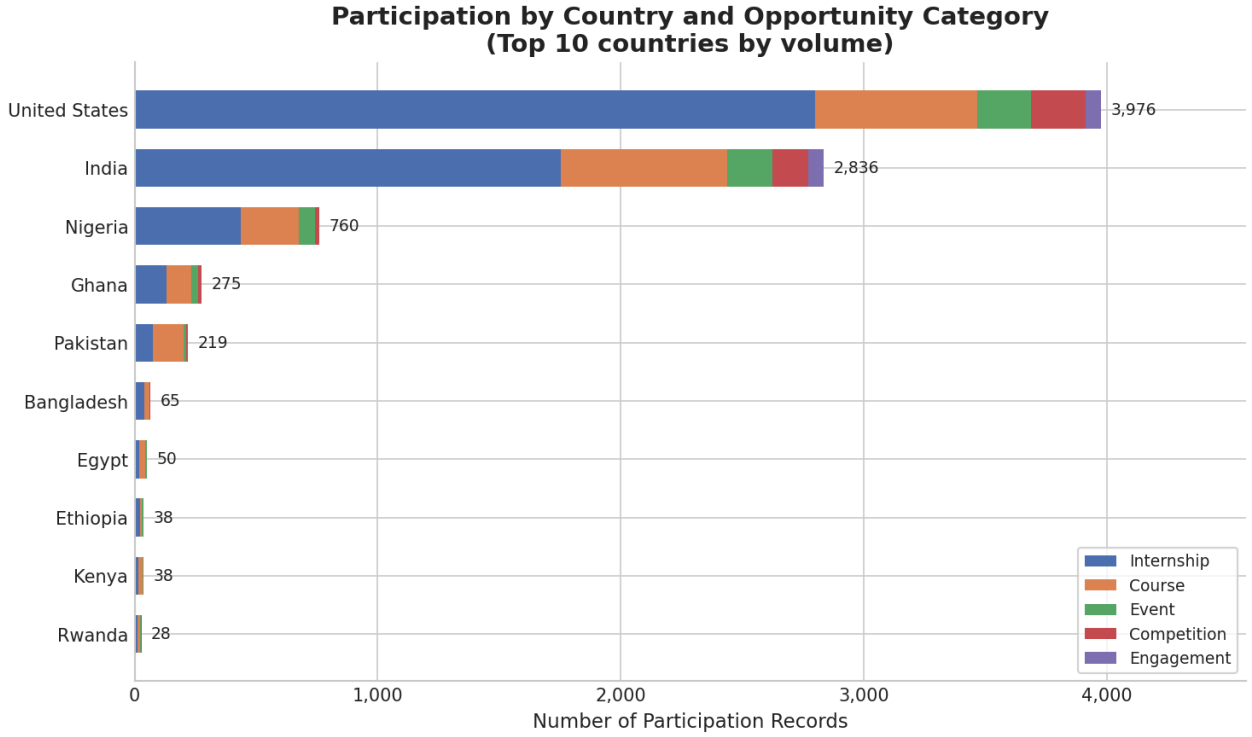


Outcome Rate by Category — this is the strongest new finding: Internships have a **78% exit rate** (Rejected/Withdraw/Dropped Out) vs. under 2% for every other category. Courses, Events, and Engagement all convert 93%+ of participants forward. This is a good candidate for a callout in your "Opportunity Category Performance" subsection, since it's currently blank in the doc.

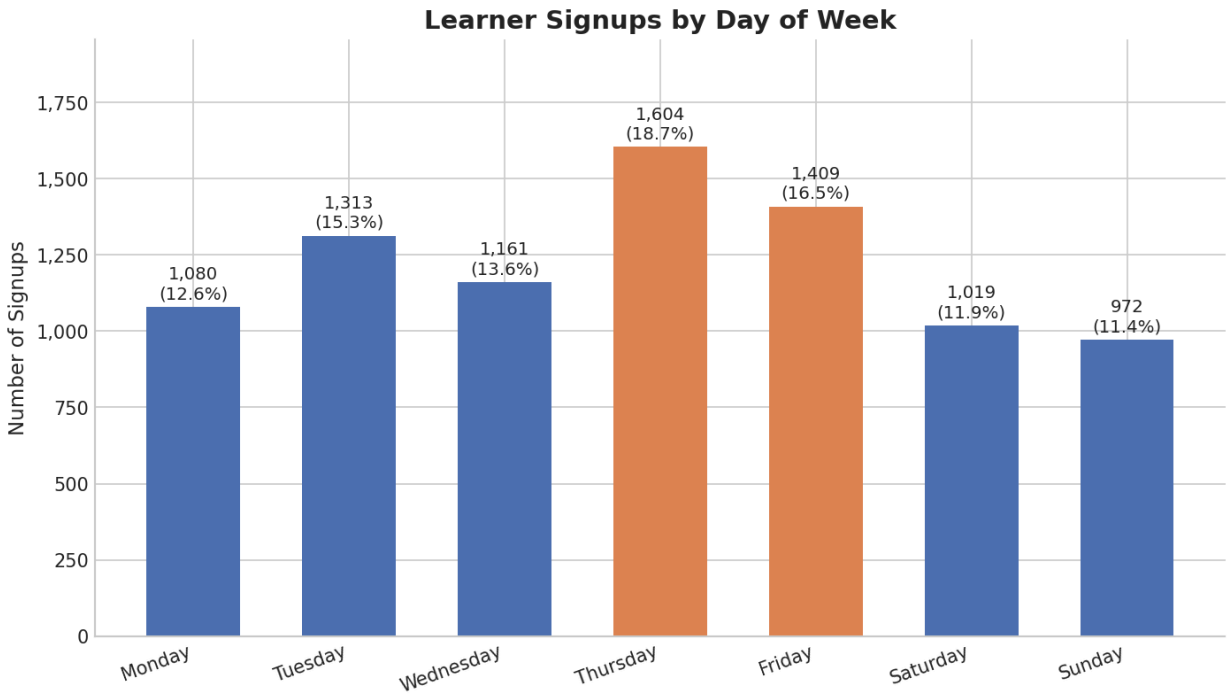


Participation by Countries and Category — Participation is heavily concentrated in two countries: the United States (3,976 records, 46.5%) and India (2,836 records, 33.1%) together account for nearly 80% of the dataset. Nigeria is a distant third (760, 8.9%), followed by Ghana (275) and Pakistan (219). Within every one of the top 10 countries, Internship is the largest single category and Course is second — the same skew seen in the overall category distribution — so the

Internship bottleneck identified earlier is not an artifact of any one country; it shows up wherever volume is high. Engagement records are essentially absent outside the US and India (only India, the US, and Nigeria have any Engagement participation at all), consistent with Engagement being the smallest and most geographically limited category.



Signup Day of Week — Addressing the original brief's question of which factors influence participation: signup activity is not evenly distributed across the week. Thursday (1,604 signups, 18.7%) and Friday (1,409, 16.5%) are the two busiest days, together accounting for over a third of all signups, while weekends are the quietest (Saturday 11.9%, Sunday 11.4%). This suggests learners are more likely to sign up during the work/school week than on weekends, with activity building toward the end of the week.



Note on engagement duration: The brief also asked about "engagement duration" as a potential driver of participation. We tested this using the gap between Apply Date and Opportunity Start Date, but it isn't a usable metric here: Opportunity Start Date is missing for 44% of records (documented in the Data Quality Report), and among the records where both dates are present, a meaningful share show the opportunity starting *before* the application was made — the same illogical date-ordering issue already flagged for validity. Rather than chart a duration measure built on data we know to be unreliable, we're flagging this as a limitation: a duration-of-engagement analysis would need cleaner start-date data from the source system before it could support a reliable conclusion.

AI-Assisted Analysis

AI Tool Used

Claude AI was used as an AI-assisted analytical tool throughout the project. It supported data quality assessment, exploratory analysis, interpretation of findings, visualization recommendations, and report development.

Prompts Used

Claude was prompted to assist with the following tasks:
• Identifying data quality issues, including missing values, corrupted timestamps, inconsistent capitalization, and illogical date sequences.
• Structuring the data quality assessment around completeness, uniqueness, consistency, validity, and accuracy.
• Identifying and summarizing historical participation trends and growth patterns.
• Comparing performance across Opportunity Categories and identifying notable differences in participation outcomes.
• Identifying potential bottlenecks and unusual patterns within the dataset.
• Recommending appropriate visualizations to communicate key findings effectively.

Insights Generated

The AI-assisted analysis helped identify and interpret several important patterns in the dataset. One notable finding was the Internship bottleneck pattern: Internship opportunities accounted for the largest share of participation records, but the proportion progressing to positive outcomes was comparatively low. This indicates that high participation volume does not necessarily translate into successful outcomes and may point to potential bottlenecks within the internship participation process.

The analysis also identified a two-wave seasonal growth pattern, with significant increases in participation around August 2023 and January 2024. These periods represent notable peaks in student opportunity participation.

In addition, the analysis highlighted category-specific status patterns. For example, the Rejected status appeared only within Internship records, indicating a distinctive outcome pattern within that category.

Validation Process

AI-generated outputs were not accepted without verification. The identified patterns, calculations, data-cleaning recommendations, and visualizations were reviewed and cross-checked against the raw dataset. Calculated figures were independently checked where necessary, and only findings supported by the underlying data were included in the final analysis.

Conclusion

Claude served as an analytical support tool rather than a replacement for human analysis. Its outputs helped accelerate pattern identification and interpretation, while the project team remained responsible for reviewing, validating, and approving the final findings.